



Advancements in *Escherichia coli* secretion systems for enhanced recombinant protein production

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Abstract

Escherichia coli is inarguably one of the most studied microorganisms across the spectrum of microbiology. It is very widely used in recombinant protein production owing to its rapid growth, ease of genetic manipulation, and relatively high protein yields. Despite all of its advantages, its inability to efficiently secrete proteins naturally remains a drawback leading to protein aggregation as inclusion bodies in the cytoplasm and consequent low overall protein yield. Therefore, many approaches to mitigate this weakness and enhance extracellular secretion to increase protein yield have been devised. This review explores the natural and engineered secretion systems in *E. coli*, highlighting their potential for enhanced protein secretion for non-glycosylated proteins. Natural one-step (e.g., Type I and III Secretion Systems) and two-step systems (e.g., Sec and Tat pathways) are detailed alongside recent advancements in genetic engineering, mutagenesis, and synthetic biology approaches aimed at improving protein yield, folding, and secretion efficiency. Emerging technologies, such as the ESETEC[®] and BacSec[®] platforms, promise scalable and cost-effective solutions for higher protein production. Challenges, including limited cellular capabilities and protein aggregation, are addressed through innovative strategies like cell wall modification, co-expression of chaperones, and medium optimization. This review emphasizes *E. coli*'s adaptability to industrial applications, and the promising future of recombinant protein technologies.

Keywords *Escherichia coli* · Recombinant proteins · Protein secretion · Protein translocation · ESETEC[®] · BacSec[®].

Abbreviations

ABC	ATP Binding Cassette	FGFR	Fibroblast Growth Factor Receptor
ATP	Adenosine Triphosphate	Ffh	Fifty-Four Homologues
BENNY	Bacterial Extracellular Protein Secretory System	GFP	Green Fluorescent Protein
BRP	Bacteriocin Release Proteins	GRAS	Generally Regarded as Safe
CGT	Cyclodextrin-glycosyltransferase	hEGF	Human Epidermal Growth Factor
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats	HlyA	Alpha Hemolysin
<i>dac</i> A and B	d, d-carboxypeptidase A and B	IM	Inner Membrane
DNA	Deoxyribonucleic Acid	LPS	Lipopolysaccharide
		MFP	Membrane Fusion Protein
		MNNG	N-Methyl-N'-nitro-N-nitrosoguanidin
		OM	Outer Membrane
		OMP	Outer Membrane Protein
		OPH	Organophosphorus Hydrolase
		OsmY	Osmotically Inducible Protein Y
		rHA	Recombinant Human Albumin
		RNA	Ribonucleic Acid
		RNC	Ribosome-bound nascent chain
		Sec	General Secretory
		SRP	Signal Recognition Particle
		TxSS	Type x Secretory System
		Tat	Twin Arginine Translocase

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Introduction

Since their introduction in the early 1970s (Jackson et al. 1972), recombinant proteins have established a strong footing in the food and drug industries. Some of the commonly used recombinant proteins in these industries include insulin (therapeutics) and β -galactosidase (dairy industry). To address their growing ubiquity, it is vital that each new generation of innovation is not only more efficient in terms of production, but also function. In the case of pharmaceuticals, it is vital to ensure that a novel generation of recombinant therapeutics is more pharmacologically efficacious, with novel modes of administration, increased specificity, improved pharmacokinetics, and the reduction of side effects. The protein of interest's characteristics- length, complexity and function have to be considered before choosing the best possible system for its production. This ensures maximum efficiency (quality, cost, and product yield) while using the most suitable production system. Based on complexity, mass production of proteins may either utilize a prokaryotic (*E. coli*) or a eukaryotic expression system (*Saccharomyces cerevisiae* and mammalian cells) (Rashid 2022). Prokaryotic systems are often used for the mass production of various proteins due to their rapid replication time, ease of genetic manipulation, and high protein yield. One of the most commonly used prokaryotic production systems is the Gram-negative organism *Escherichia coli*.

Escherichia coli is one of the most widely used organisms for the production of recombinant proteins in industry. This can be attributed to the organism's ability to achieve high cell numbers in a low amount of time using by relatively cheap energy sources. Various strains of *E. coli* exist for production. They have been designed for ease of genetic manipulation and are Generally Regarded as Safe (GRAS) (Fijan 2014; İncir and Kaplan 2024). Despite being a great host for recombinant protein production, *E. coli* still possesses various drawbacks; it fails to provide various post-translational modifications to a lot of recombinant proteins such as the efficient folding of proteins, glycosylation, methionine modification, and creation of di-sulphide bridges (Bhatwa et al. 2021). It is also a poor protein secretor, hence leading to recombinant protein accumulation in the cytoplasm as inclusion bodies leading to reduced yield of desired proteins (Sadeghian-Rizi et al. 2019). *E. coli* naturally secretes proteins through two broadly categorized systems of secretion – The one-step secretion system, and the two-step secretion system. In contrast to methods like cell lysis, using these secretory systems comes with various advantages like ease of isolation for non-glycosylated proteins (Freudl 2018; Ahan et al. 2023), low risk of proteolytic degradation without host cell protein contamination. Therefore, these systems are optimized for enhanced secretion

through techniques such as genetic engineering, medium optimization, and process optimization, with demonstrably successful results (Karyolaimos et al. 2019a; Chen et al. 2021; Yang et al. 2022; Ahn and Jung 2023).

Structure of *Escherichia coli* membrane

Since *Escherichia coli* is Gram-negative, its cell envelope consists of three layers – the Inner Membrane (IM), a thin network of peptidoglycan, and an Outer Membrane (OM) (Fig. 1a). The IM is composed of a phospholipid bilayer that is selectively permeable (Pluhackova and Horner 2021; Weiner and Li 2008) and allows the translocation of proteins and other metabolites essential for lipid synthesis, pathogenesis and secretion. This membrane is the primary site of protein translocation via the Sec and Tat pathways, which transport unfolded and folded proteins, respectively, into the periplasm for further processing (Kudva et al. 2013). The peptidoglycan layer lies in the space between the IM and OM and is essential to provide strength, support and structural integrity to the organism (de Pedro and Cava 2015). The space between the OM and the IM is composed of a viscous substance called the periplasm (Kiss et al. 2021). The periplasm provides a highly oxidative environment that facilitates protein folding, disulfide bond formation, and chaperone-assisted stabilization, all of which are critical for effective secretion (Goemans et al. 2014). Proteins transported via the Sec and Tat pathways often undergo modifications in the periplasm before being translocated to the extracellular space. However, periplasmic accumulation can sometimes result in protein aggregation, necessitating further optimization of secretion pathways (Baneyx and Mujacic 2004).

Outer Membrane Proteins (OMPs) typically come with a β -barrel structure (Schulz 2002; Orme et al. 2006; Weiner and Li 2008) whose main functions are enzymatic reactions and translocation. The bilayer is composed of phospholipids on the inside, and lipopolysaccharides (LPS) on the outside. This asymmetrical bilayer is responsible for the protection of the bacterial cell from the external environment (Kiss et al. 2021). The lipopolysaccharides in the outer layer can evoke a severe immune response in host organisms through their constituent endotoxins that are released when the bacterium is lysed (Dufour et al. 2017). The OMPs act as transporters or enzymes that help in secretion of various proteins, including certain virulence causing factors and biofilm forming molecules (Orme et al. 2006; Sora et al. 2021; Pokharel et al. 2023). OMPs such as TolC in the Type I Secretion System (TISS) facilitate direct protein transport from the cytoplasm to the extracellular environment (Alvarez-Martinez et al. 2021). The asymmetry of the OM, with LPS on its exterior, contributes to bacterial defense but also

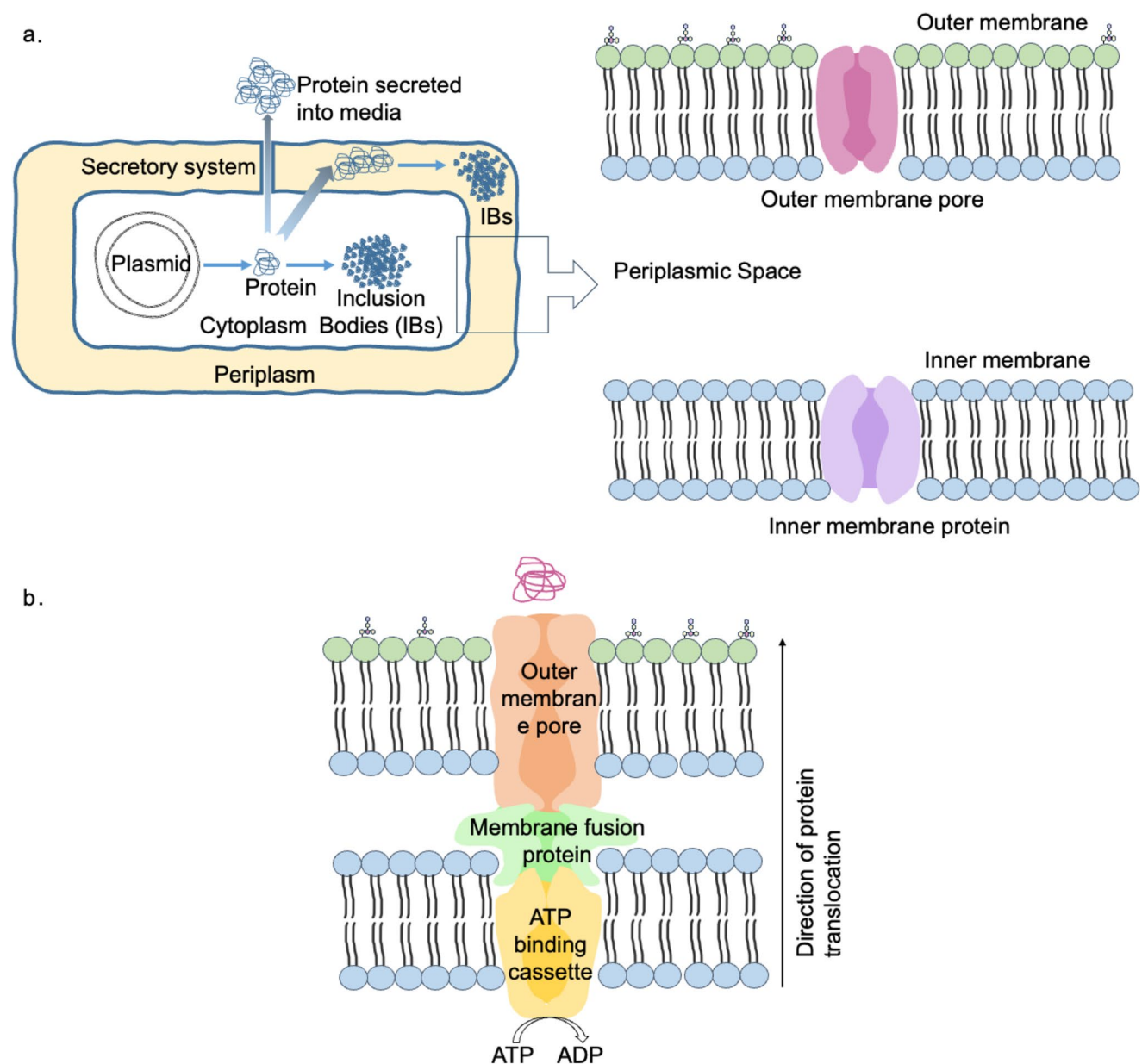


Fig. 1 The structural elucidation of *Escherichia coli* membranes and One step secretion system. **(a)** This provides us a magnified overview of the *E. coli* cell wall structure and demonstrates how it is a naturally poor secretor. Typically, a significant part of the generated recombinant proteins aggregates in the cytoplasm and the periplasm as inclusion bodies (IB), in avoidance of proteolytic cleavage, with the rest

getting secreted into the media extracellularly. **(b)** The Type I Secretion System or TISS, a type of one-step secretion system, uses the ATP Binding Cassette (ABC), The Membrane Fusion Protein (MFP), and The Outer Membrane Pore (OMP) to form a channel bypassing the periplasm entirely, helping the protein to directly translocate extracellularly from the cytoplasm

presents challenges for recombinant protein secretion (Kaur and Mingeot -Leclercq 2024). Additionally, outer membrane porins and β -barrel structures contribute to secretion pathways by enabling selective translocation of proteins (Otzen and Andersen 2013). These membrane proteins are vital components of multi-protein complexes involved in secretion, including the Type II Secretion System (T2SS) and Type V Secretion System (T5SS), which are responsible for transporting proteins from the periplasm across

the OM (Nivaskumar and Francetic 2014; Meuskens et al. 2019). Modifying these systems through synthetic biology approaches has shown promise in improving extracellular protein yields.

Finally, the cell envelope serves as a site of attachment for appendages like flagella and fimbriae that help in bacterial cell motility, communication and attachment (Jin and Marshall 2020; Kreve and Reis 2021). The membrane proteins present in the cell envelope, as stated above, play

an important role in bacterial protein secretion (Rego et al. 2010; Goemans et al. 2014; Kleiner-Grote et al. 2018; Karyolaimos et al. 2020). They are involved in both the one step and two step secretion mechanisms. In their natural state, these secretion systems are used for the secretion of molecules that are responsible for pathogenicity (toxins) and organism survival (Hinsa et al. 2003; Rego et al. 2010; Costa et al. 2015; Smith et al. 2018). Thus, understanding and engineering the *E. coli* membrane structure is crucial for optimizing recombinant protein secretion, whether by enhancing membrane permeability, modulating secretion machinery, or employing secretion signal sequences to improve translocation efficiency.

The natural secretion systems of *Escherichia coli*

The following sections will elucidate the secretory mechanisms of *Escherichia coli* and how these mechanisms can be engineered and modified to enhance recombinant protein production.

The one step secretion system

Since *E. coli* is genetically malleable, various non-native systems of secretion can also be leveraged alongside its natural secretion systems. Typically, *E. coli* uses the T1SS (Type I Secretion System) and, along with the flagellar T3SS (Type III Secretion System), which has evolved into a syringe-like structure, for protein translocation in its natural state (Filloux 2022).

The Type 1 Secretion System or T1SS (Fig. 1b) has been defined as a protein translocation system that utilizes three membrane proteins for its function – an inner membrane ATP Binding Cassette (ABC), an Outer Membrane Pore (OMP) and a Membrane Fusion Protein (MFP) (Yoon et al. 2010; Smith et al. 2018; Kanonenberg et al. 2018; Pourhassan N et al. 2022a, b). These three proteins form a channel between the cytoplasm and the cell's exterior, allowing the protein to bypass the periplasm entirely (Thomas et al. 2014; Zou et al. 2021). The Type 1 Secretion System is highly versatile and is used by the organism to transfer peptides as small as < 10 kDa or massive proteins with mass as big as 1 MDa (Thomas et al. 2014; Smith et al. 2018). T1SS is a highly conserved secretion mechanism across a multitude of organisms and it is vital for bacterial survival and adaptation (Li et al. 2023). In its natural state, the T1SS translocate the adhesins, toxins and proteases that are useful for multiple functions important to the cell ranging from pathological responses in host organisms to biofilm formation in the external environment (Qin et al. 2022; Carezzano et al. 2023). One example for the utilization of the Type 1

Secretion System naturally is the secretion of α - Hemolysin (Gentschev et al. 2002).

The secretion of Hemolysin A utilizes the ABC protein HlyB, the MFP HlyD, and the OMP TolC (Thomas et al. 2014; Pourhassan et al. 2023). HlyA is very commonly used as the transporter molecule when utilizing the one-step secretion system for production of a recombinant protein of choice. The recombinant protein interacts with the C-terminal region of HlyA which leads to HlyB and HlyD forming a complex. The HlyB/HlyD complex binds to HlyA attached to the target protein in the cytoplasm. TolC interacts with this HlyB/HlyD complex and HlyA + protein to form a channel allowing the HlyA and protein to bypass the periplasm and exit the cell, leading to the detachment of the two membranes (Spitz et al. 2022; Pourhassan N et al. 2022a, b). Other proteins that can be secreted by the T1SS include toxins, enzymes like proteases and lipases and adhesins. The molecular weights of these proteins can be between 19 kDa and 900 kDa (Kim et al. 2023). However, there are several drawbacks of utilizing the one step secretion system for recombinant protein production. The target protein of choice requires a vehicle for secretion (Ex: Protein + HlyA), requiring more resources for further processing i.e. the removal of the transporter HlyA, for instance; the secretion system depends on the co-expression of HlyB/HlyD for secretion and since this is highly conserved mechanism of protein translocation, microbes often utilize it to secrete their own proteins, not just the recombinant proteins (Narayanan et al. 2010; Pourhassan et al. 2023). Therefore, it may contribute to a comparatively low yield of protein of interest. To tackle these challenges, induced co-expression of HlyB/HlyD along with addition of a protease cleavage site between the protein of interest and carrier HlyA have been suggested as methods to increase protein yields and reduce further downstream costs. Alongside the above stated optimization methods, various other one step secretion systems and possible modifications have also been looked into (Pourhassan N et al. 2022a, b). Numerous proteins have been successfully secreted by utilizing the HlyA T1SS including the enzymes β -galactosidase (Pourhassan N et al. 2022a, b) and cutinase (Su et al. 2012). Many recombinant proteins are produced as fusion proteins to enhance solubility or stability. These are cleaved using targeted proteases such as enterokinase or TeV protease, enabling precise removal of fusion tags with minimal impact on the protein's functionality. It has also been seen that by utilizing this method, modified proteins i.e., proteins with di-sulphide bridges, dimeric and tetrameric proteins can be successfully secreted (Fernandez and de Lorenzo 2001). Along with *E. coli*'s native secretory systems, various other heterologous secretory systems can be engineered into the organism to enhance protein secretion. Examples include the secretory systems of *Pseudomonas*

fluorescens (Idei et al. 1999), and *Bacillus subtilis* (Eom et al. 2016).

Along with the T1SS, T3SS is also involved in the protein translocation between the cytoplasm and the external medium (Wagner et al. 2018; Miletic et al. 2021). The Type 3 Secretion System (T3SS) is a highly specialized secretion pathway predominantly found in Gram-negative bacteria. It functions as a molecular syringe, injecting effector proteins directly into host cells to manipulate their cellular processes. This system is composed of a multi-protein complex that spans the bacterial membranes and extends outward as a needle-like structure, which provides a direct translocation pathway bypassing the periplasmic space (Coburn et al. 2007). Although primarily associated with pathogenicity, T3SS has gained increasing interest in biotechnology due to its potential for targeted protein delivery. Researchers have explored its use in recombinant protein secretion by modifying effector recognition signals, adapting secretion chaperones, and engineering non-pathogenic bacterial strains to exploit this system for biotechnological applications (Diepold and Armitage 2015; Bai et al. 2018). Unlike T1SS, which facilitates broad extracellular secretion, T3SS enables intracellular delivery of proteins, making it a promising tool for applications in therapeutic protein delivery, vaccine development, and synthetic biology. The Flagellar evolved T3SS can be used for the transfer of proteins by making certain modifications to the organism (Young et al. 1999). For the purpose of secretion of recombinant proteins, the strain *E. coli* MKS12 was engineered to lack the ability of producing fimbriae and flagella by deleting genes such as *fliD* and *fliC*. The deletion of both these genes together allowed the secretion of proteins directly from the cytoplasm to the exterior of the cell (Young et al. 1999; Anton et al. 2010). One such protein secreted successfully using this method is α -enolase from *Staphylococcus pneumoniae* (Majander et al. 2005).

Efforts are being made to enhance protein secretion efficiency by optimizing the expression of T3SS-associated chaperones and secretion signals, which could broaden its utility in industrial protein production (Pendergrass and May 2020). Despite these advancements, challenges remain in adapting T3SS for large-scale recombinant protein production. The requirement for host cell interaction, limited substrate specificity, and relatively lower secretion yields compared to T1SS and T2SS present significant hurdles. However, ongoing research into synthetic biology approaches, including genome editing and pathway reconstitution in non-pathogenic strains, holds promise for overcoming these limitations (Zhou et al. 2014; Reeves et al. 2015; Liang et al. 2022). As further developments in T3SS engineering continue, its role in recombinant protein production may expand, making it a viable alternative for

specialized applications where intracellular protein delivery is beneficial (Reeves et al. 2015; Gershberg et al. 2024). While its sophisticated needle-like structure has potential applications in targeted protein delivery, its relevance to large-scale recombinant protein secretion is limited compared to T1SS. However, recent studies have explored engineering T3SS for non-pathogenic applications, such as delivering therapeutic proteins or modifying secretion specificity (Lindner et al. 2020; Lim et al. 2020). If further advancements make T3SS a viable option for recombinant protein production, an expanded discussion of its mechanisms and engineering strategies would be warranted.

In addition to the well-characterized secretion systems, certain protein secretion pathways remain uncategorized and largely unexplored. However, computational advancements have provided a promising avenue for uncovering these elusive mechanisms by leveraging the genetic blueprint of bacteria. The bacterial genome holds vital information that can illuminate the diverse routes taken by proteins during secretion. Cutting-edge computational tools have been developed to predict protein secretion systems and their associated appendages, particularly in bacteria with lipopolysaccharide (LPS)-containing outer membranes. These comprehensive frameworks have enabled the identification of novel secretion systems, such as T6SSiii and T9SS, which are uniquely restricted to specific bacterial clades, offering exciting insights into bacterial physiology and evolution (Abby et al. 2016, 2024).

The two step secretory system

Another approach that is utilized for the translocation of proteins in *E. coli* is the two-step secretory system. This system of translocation involves the secretion of the proteins first through the IM and then through the OM including passage through the periplasm unlike the one-step system (Yu et al. 2023). The two pathways well known to adapt the two-step secretion are the General Secretory (Sec) (R and Joan 2016; Crane and Randall 2017) and the Twin Arginine Translocase (Tat) (Lee et al. 2006; R and Joan 2016) pathway. Both these pathways, though two-step, have fundamental differences in the structure of the proteins transferred (Natale et al. 2008; Fröbel et al. 2012a; Crane and Randall 2017). Since *E. coli* is naturally a poor secretor, using the two-step secretion mechanism allows for the enhanced secretion of numerous proteins. Modifying the outer membrane allows for the enhanced release of proteins from within the periplasm to the cell exterior and to that end, various strategies have been successfully implemented.

The translocation of protein via inner membrane

In this two-step secretory system, based on state of the protein, either the general secretory (Sec) or the Twin Arginine Translocase (Tat) pathway can be employed for the translocation of the protein across the IM into the periplasm.

The general secretory (Sec) pathway

In the Sec-pathway, the target proteins are coupled with SecA/SecB (Lycklama and Driessen 2012) or signal recognition protein (SRP) (Saraogi and Shan 2011). With the help of a signal peptide, many proteins enter Sec translocase present on the IM and translocate the proteins into the periplasm (Fig. 2a). The Sec translocase, which typically includes the proteins SecYEG and SecA is used for the post-translational secretion of proteins. SecA is an ATP-dependent motor (Tsirigotaki et al. 2017; Gupta et al. 2020) and while SecYEG complex forms a heterotrimeric protein conduction channel for the translocation of proteins across the IM into the periplasm (Sachelaru et al. 2017), SecB stabilizes the unfolded proteins before they are translocated via the Sec pathway (Eismann et al. 2022). Unlike the SecA/SecB dependent pathway, the proteins undergo modification as they are translocated i.e., proteins that are being co-translated, are dependent on the Signal Recognition Proteins (SRP). The SRP consists of fifty-four homologs (Ffh), 4.5 S RNA, and the membrane bound receptor FtsY (Pross and Kuhn 2019). The SRPs binds to the ribosome-bound nascent chain (RNC) and targets it to FtsY, where the SRP bound RNC is sent to the Sec translocase (Sachelaru et al. 2017). The SecB dependent and the SRP protein translocation occurs competitively and is based on the various factors like the hydrophobicity of the protein N-terminal signal sequence and the binding of the signal factors (Beck et al. 2000; Lee and Bernstein 2001; Halic et al. 2004). The translocation capacity of the general secretory pathways is limited and can lead to cytoplasmic protein accumulation due to its inability to cope with the load. Therefore, inclusion body formation and proteolysis is the reported consequence (Jong et al. 2017). To bypass this, secretion can be enhanced by amplifying the genes involved in the Sec dependent pathways like SecYEG (Oswald et al. 2021), SecA (Zhu et al. 2022). It has also been observed that impact on translocation can be seen based on type of protein translocated (Berges et al. 1996).

The twin arginine translocation (Tat) pathway

Folded proteins (Taubert et al. 2015), proteins with co-factors (Palmer and Berks 2012) are translocated across the IM via another pathway known as the Twin Arginine Translocase Pathway or the Tat Pathway (Fig. 2b). The Tat pathway

utilizes the complex of TatBC and Tat (A) TatBC recognizes a specific signal and binds to it. This results in the recruitment of various TatA monomers that form a transmembrane channel for protein translocation. Proton Motive Force or PMF provides the energy for the translocation across the IM into the periplasm (Berks et al. 2000; Hamsanathan and Musser 2018; Alcock and Berks 2022). Along with TatA, a similar protein called TatE has also been observed, but in much smaller quantities (Eimer et al. 2015). Examples of recombinant proteins translocated using the Tat system include glucose-fructose oxidoreductase (Lindenstrauss and Bruser 2006), green fluorescent protein (Li et al. 2009), and alkaline phosphatase (DeLisa et al. 2003). Unlike the Sec system, the Tat translocons are expressed very minimally (Frain et al. 2019). Therefore, their scope of usage is comparatively very limited. But, due to its ability to only transfer folded proteins, protein accumulation within the cytoplasm was found to be significantly reduced while using this system for protein production. Hence, to solve this problem, overexpression of Tat translocons has been devised as a solution (Karyolaimos and de Gier 2021). For example, the co-expression of non-native Tat translocons of (*B. subtilis*) (Pop et al. 2002), *Rhodobacter capsulatus* (Li et al. 2009) and have already been employed successfully for translocation of target protein in *E. coli*. The Tat translocation is inefficient compared to Sec translocation due to the lower expression of these systems when compared to the latter (Natale et al. 2008). Co-expression of chaperones seem to boost recombinant proteins translocation efficiency (Muminov et al. 2023). For example, increased translocation was observed when co-expression of the chaperone of TorD and TorA signal sequence were used for the secretion of GFP protein. The signal sequence is protected further by degradation due to interaction with TorD (Kleiner-Grote et al. 2018).

Protein transport through outer membrane

The General Secretory (Sec) Pathway and The Twin Arginine Translocase (Tat) Pathway are used for the translocation of the proteins in unfolded and folded states, respectively, across the inner membrane to the periplasmic space (Frain et al. 2019; Allen and Collinson 2023). The translocation across the outer membrane to the external environment occurs via the Type 2 Secretion System (T2SS) and The Type 5 Secretion Systems (T5SS) (Korotkov et al. 2012).

The type 2 secretory system

Originally thought to be the terminal branch of the Sec pathway but also discovered to be of the Tat's, the Type 2 Secretory System is a conserved system across various

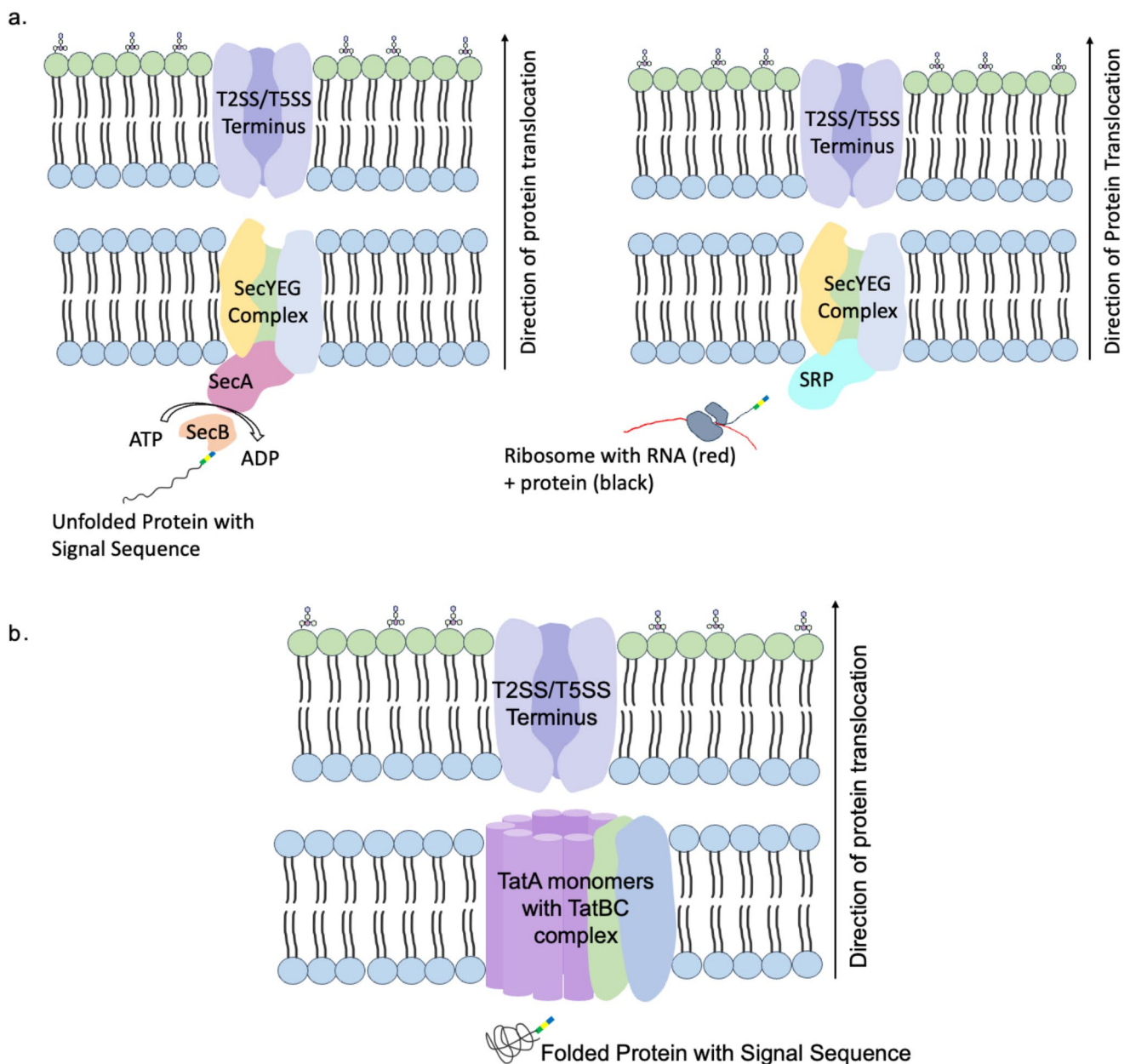


Fig. 2 The Sec and Tat pathway. **(a)** the Sec system exclusively secretes unfolded proteins across the inner membrane. The complete unfolded protein utilises the SecA and B molecules and actively translocates across SecYEG. The co-translational secretion of proteins on the other hand depends entirely on the SRP and FtsY to pass through the SecYEG complex. The extracellular secretion of these proteins from the periplasm can either occur through the T2SS or the T5SS. **(b)**

The Tat pathway or the twin arginine translocase pathway exclusively secretes folded proteins across the inner membrane into the periplasmic space, the folded protein along with the signalling sequence binds to the TatBC complex which gathers TatA monomers that form a channel allowing the proteins to go across the inner membrane into the periplasm. Like the Sec pathway, the Tat pathway also uses either the T2SS or the T5SS to secrete the protein extracellularly

Gram-negative bacterial species (Shaliutina-Loginova et al. 2023). Its main function is to transport folded proteins from the periplasmic space to the external environment through the outer membrane. The protein in question has to be secreted into the periplasmic space either through the General Secretory (Sec) or the Twin Arginine Translocase (Tat) pathway (Voulhoux et al. 2001; Cianciotto 2005).

Therefore, the machinery should have N-terminal signal sequences competent for proteins that are transferred via both these pathways (Palmer and Stansfeld 2020). Both the Sec and Tat system utilise the T2SS. While the protein that are translocated via the Sec system undergo modifications in the periplasm before extracellular translocation, the Tat system in contrast translocates proteins that have been

folded and modified (Pei and Dalbey 2022). The T2SS has a broad specificity and is capable of secreting a diverse range of molecules across the outer membrane. This enables the system to secrete a single protein and to also secrete various proteins based on the species of bacteria (Korotkov et al. 2012; R and Joan 2016). These proteins have a broad range of functions that are important for the survival of the bacterial species that they are present in, but mostly, the T2SS is used for the transport of enzymes like lipases, phosphatases, proteases, and enzymes that process various carbohydrates (Chimalapati et al. 2020; Korotkov et al. 2012).

The type 5 secretory system

The T5SS is characterised by the simplest and the most widespread of all the secretory systems among the Gram-negative bacteria. Autotransporters, the most well-known of T5SS (Pohlner et al. 1987) contain both their own translocator and the cargo needed for translocation, making it one of the most efficient translocation systems in Gram-negative bacteria (Leyton et al. 2012; Irene et al. 2023). Like T2SS, the T5SS also transports proteins from the periplasm to the exterior of the bacterial cell (Leyton et al. 2012). The protein that the secretory system carries is first secreted either by the Sec or the Tat pathway. Though the T5SS autotransporters are the most widely known variety of T5SS, there are more varieties of T5SS in existence – like the two-partner secretion and the chaperone usher pathways (van Ulsen et al. 2014; Dautin 2021).

In the two-partner secretion pathway of the T5SS, two proteins participate in the secretion mechanism. One carries the β -barrel while the other is the protein that needs translocation. This method of secretion is typically observed for the secretion of proteins of large molecular weights that typically cause virulence like in *Bordetella pertussis* (Dautin 2021).

The chaperone usher pathway, a constituent mechanism of the Type 5 Secretory System, is involved in the two-step protein secretion. In this usher pathway, two proteins are involved to translocate the protein from the periplasm to the exterior of the cell via the outer membrane. The two proteins – the periplasmic chaperone and the usher proteins are used for folding the protein and forming the β barrel for the protein translocation (Balchin et al. 2020; Huynh et al. 2024). The chaperone protein forms the molecular scaffold for the assembling and correct folding of proteins called the usher proteins. The chaperone-usher dimer recruits even more usher proteins to form a pilus on the outer membrane surface. The chaperone usher pathway is well regarded for its role in uropathogenesis and being looked into as a potential novel antibacterial therapeutic target (Whelan et al. 2023; Stewart and Schisler 2024).

As recombinant protein production shifts from producing whole molecules to active parts such as antibody fragments (e.g., Fab, scFv, or nanobodies), several challenges emerge in secretory secretion systems. A significant issue lies in folding and assembly, as antibody fragments require proper disulfide bond formation and folding, which occur in the oxidative environment of the periplasm (Bruno et al. 2019). The Sec pathway, commonly used for protein translocation, transports unfolded proteins, increasing the risk of misfolding or aggregation in the periplasm. Misfolded antibody fragments not only compromise functionality but may also burden the host cell with proteotoxic stress (Ajmal 2023). Smaller fragments like nanobodies or single-chain variable fragments (scFvs) face an additional challenge as their size and structure might fail to efficiently engage the secretion machinery, leading to suboptimal yields (R and Joan 2016). Furthermore, secretion systems often lack specificity for these smaller or engineered fragments, and their hydrophobicity or conformational requirements can further impede efficient transport. In addition, antibody fragments can be highly susceptible to proteolytic degradation, especially in the periplasm or extracellular environment, due to their exposed structure or lack of stabilizing domains found in full antibodies (Karyolaimos and de Gier 2021). Engineering host cells with enhanced chaperones, oxidoreductases (e.g., DsbA, DsbC), or protease-deficient strains can mitigate these challenges, but such modifications add complexity to the production process. The energy-intensive nature of secretion pathways like Tat or Type II further complicates scalability, as these pathways may struggle to meet the demands of industrial-scale production (Palmer and Berks 2012). Moreover, downstream processes like purification and refolding become more critical for these smaller fragments, as improper secretion or incomplete folding directly impacts their therapeutic efficacy. To address these issues, signal peptide optimization, host strain engineering, and synthetic biology approaches can be employed to tailor secretion systems specifically for antibody fragments. However, optimizing the balance between secretion efficiency, yield, and functional quality remains an ongoing challenge in this evolving landscape of recombinant protein production.

Secretion signal sequences useful for Recombinant protein translocation

A lot of decisions have to be taken into account before using *Escherichia coli* as a secretion system for a recombinant protein of choice. The typical decisions include – protein characteristics and the strain used for its production, but, of equal importance are the signal sequences that help in directing the protein of choice to the Sec/Tat translocation

systems (Pang et al. 2022; Hashemzaei et al. 2023). Optimizing signal peptides enhances recombinant protein secretion by improving translocation efficiency, stability, and specificity. Key strategies include rational design and mutagenesis to modify hydrophobicity and cleavage sites for better secretion, fusion with carrier proteins like MBP or OsmY to stabilize nascent polypeptides and prevent degradation (Pennati et al. 2014; Gonzalez-Perez et al. 2021a), and screening through directed evolution to identify high-efficiency signal peptides via genetic libraries (Heggeset et al. 2013; Alias et al. 2022). Additionally, synthetic signal peptides can be designed using computational tools like SignalP and TatP, while strain engineering approaches, such as co-expressing chaperones or modifying membrane permeability, further enhance secretion efficiency (Bendtsen et al. 2005; Teufel et al. 2022).

The sec and SRP pathway signal sequences

The Sec (General Secretory) pathway primarily facilitates the translocation of unfolded proteins across the inner membrane into the periplasm and is the most widely utilized secretion system due to its high efficiency and broad substrate specificity (Driessen and Nouwen 2008). It consists of key components such as SecA (an ATPase motor), SecYEG (the translocon), and SecB (a chaperone that stabilizes precursor proteins before translocation). The energy required for translocation is provided by ATP hydrolysis and the proton motive force (Lycklama and Driessen 2012; Ghaisani and M 2019). This system is highly efficient and well-characterized for recombinant protein production, capable of transporting a broad range of proteins, including large and complex molecules. It is also compatible with the co-expression of chaperones to enhance proper folding post-translocation. However, the Sec pathway translocates proteins in an unfolded state, necessitating additional folding machinery in the periplasm. This process makes proteins susceptible to aggregation within the periplasm, potentially lowering yields, and increases the risk of proteolytic degradation due to prolonged exposure in the periplasm (Natale et al. 2008; Crane and Randall 2017; Tsigotaki et al. 2017).

A typical signal sequence for the Sec translocase, with an average of 20 amino acids in length, consists of three regions – the N, H and C regions. The N- region consists of positively charged amino acids while the H-region, represents the core of the signal sequence, therefore its length and hydrophobicity have influence on the translocation efficiency of a theoretical protein of choice (Kaushik et al. 2022). The C-region of the signal sequence consists of conserved recognition motifs with small and uncharged amino acids in positions –1 and –3 relative to the cleavage site (Paetzel et al. 2002). It has been observed that improving one region

to compensate for the deficiency of another region has an effect on the secretion efficiency (Hikita and Mizushima 1992; Karyolaimos et al. 2019b). Several homologous and heterologous signalling sequences have been successfully used for the targeting of proteins into the Sec pathway secretion machinery in *E. coli*. Homologous Sec signal sequences include LamB (Pratap and Dikshit 1998), OmpA (Thie et al. 2008; Pechsrichuang et al. 2016), OmpC (Wang et al. 2016), OmpF (Forster et al. 2013), and MalE (Sommer et al. 2009). The homologous SRT dependent pathway signal sequences include DsbA (Schierle et al. 2003; Durrani et al. 2015) and TolB (Huber et al. 2005; Steiner et al. 2006). Heterologous signalling sequences include those that have been taken from *Bordetella pertussis* (Oliver et al. 2003), *Bacillus* sp. (Low et al. 2012). Other than the ones mentioned above, there are many other known signal sequences for Sec and SRP dependent signalling sequences exist and many more are yet to be discovered.

The twin arginine translocase or Tat pathway signal sequences

The Tat (Twin-Arginine Translocation) pathway, in contrast to Sec pathway is specialized for the translocation of fully folded proteins, including those with cofactors or disulfide bonds (Gallego-Parrilla et al. 2024). It operates through a distinct mechanism that involves the TatA, TatB, and TatC proteins, utilizing the proton motive force without ATP hydrolysis (Natale et al. 2004, 2008). This pathway facilitates the secretion of fully folded and functional proteins, reducing misfolding issues. It is ideal for proteins that require cofactor incorporation prior to secretion and reduces periplasmic stress since proteins are secreted in their final conformation. However, the Tat pathway has limited translocation capacity compared to the Sec pathway, making it less suitable for high-yield applications. It also requires specific twin-arginine signal sequences, reducing its general applicability, and is not as widely utilized or engineered as the Sec pathway, leading to fewer available optimization strategies (Berks et al. 2000; Fröbel et al. 2012b; Palmer and Berks 2012).

Like the Sec pathway, the Tat pathway needs signalling sequences to direct the protein of choice into the translocation machinery in the Inner Membrane (IM). The Tat dependent signal sequences like the Sec sequences are also composed of an N-, H-, and C- regions. But the signalling sequences in the Tat pathway are variable in length (25–50 amino acids) due to the differences in the length typically observed in the N-region (Ulfig et al. 2017; Frain et al. 2019). The main characteristics of the signal sequence is the arginine containing Z-R-R-X-Φ-Φ motif in the highly conserved N-region where The R stands for Arginine, Z for

a polar region, Φ for hydrophobic residues, and X for any residue (Bruser 2007; Gallego-Parrilla et al. 2024). However, there are some exceptions to this rule, the consensus remains the above. Unlike the signalling sequences of the Sec pathway, the Tat pathway's signalling sequences are less hydrophobic and less specific (Natale et al. 2008) i.e., if the proteins are unfolded, they are translocated via the Sec pathway. Factors like net charge and hydrophobicity have an impact on translocation efficiency (Wang et al. 2017; Bui and Hoang 2021; Bartoš et al. 2024). There are about 29 (known) homologous signalling sequences known for the Tat translocation system in *E. coli* (Li et al. 2009). The copper oxidase (CueO) signalling sequence has the highest resemblance to the characteristic Z-R-R-X- Φ - Φ motif (Fröbel et al. 2012a). The most commonly used signalling sequence for the expression of recombinant proteins is the TorA sequence (Kreutzenbeck et al. 2007; Bagherinejad et al. 2016). The proteins produced using TorA include rHSA (Recombinant Human Serum Albumin), hEGF (Human Epidermal Growth Factor). Heterologous translocation systems successfully used for the production of recombinant proteins include Organophosphorus Hydrolase (OPH) from *Flavobacterium* (Kang et al. 2016) and DagA from *Streptococcus* (Li et al. 2009).

Many homologous and heterologous signalling sequences exist for both Sec and Tat pathways. Most of them have been successfully used and implemented for the production of recombinant proteins. Other than those that come naturally, other approaches have been adopted to enhance efficiency through the modification of the signalling sequences. These include mutagenesis (Alias et al. 2022; Yu et al. 2022) and the development of synthetic signal peptides (Clérico et al. 2008; Johnsten et al. 2022; Umar et al. 2025).

To enhance the practical utility of Tat-dependent secretion, researchers can leverage bioinformatics tools and signal peptide databases to identify and design optimal signal sequences. Accessible resources include SignalP (Nielsen 2017), a widely used tool for predicting signal peptides and determining their likelihood of directing proteins to Sec or Tat pathways; TatP, a specialized prediction tool for identifying potential Tat pathway signal peptides in bacterial proteins; and the Expasy Signal Peptide Database, a curated collection of experimentally validated signal peptides that can serve as references for selecting effective sequences. Using these tools, researchers can improve the efficiency of Tat-dependent secretion by selecting or engineering signal peptides with enhanced specificity and translocation efficiency. Additionally, synthetic biology approaches, such as rational mutagenesis, can be employed to fine-tune Tat signal sequences, further optimizing their use in recombinant protein production. By incorporating these computational and experimental strategies, the practical challenges

associated with Tat signal sequence variability can be mitigated, allowing for more reliable and efficient protein secretion in biotechnological applications.

The choice between the Sec and Tat pathways depends on the nature of the target protein and the desired production outcomes. For high-yield production of recombinant proteins, the Sec pathway is generally preferred due to its robustness and efficiency (Crane and Randall 2017; Ojima-Kato 2025). However, for complex proteins requiring proper folding and post-translational modifications, the Tat pathway may be more suitable (Sargent et al. 2006). Engineering efforts continue to optimize both pathways, including modifying signal sequences, enhancing pathway expression, and improving host strain capabilities to accommodate a broader range of proteins. By leveraging the strengths of both pathways and integrating synthetic biology approaches, researchers can enhance recombinant protein production in *E. coli* while overcoming existing secretion challenges.

Enhancing protein translocation

As *E. coli* is known to be a good host for developing proteins but poor secretor of the same, many methods have been employed to enhance protein secretion and increase protein yield. As seen above, genetically modifying the organism to co-express different secretion pathways together has shown varying degrees of success. The following section focuses on the translocation of proteins from the periplasm to the cell's exterior i.e. the growth medium. To enhance and optimize the translocation of proteins across the outer membrane from the periplasm, various approaches already exist - mainly through the enhancement of cell wall permeability. These enhancements may be permanent (through genetic engineering), or physical, or chemical (enhancement of the growth medium) to allow the increase of protein yield.

Permanent enhanced permeability – through mutagenesis

To allow the efficient release of periplasmic proteins, many strains have been genetically developed and some strains patented (Table 1). These genetically modified and patented strains are successfully leaked the periplasmic proteins. One of the strains, DS410 (Böck August et al. 1989), is mutated to leak using N-Methyl-N'-nitro-N-nitrosoguanidin (MNNG) combined with D-cycloserin. The strain selection is promoted by the indicator cyclodextrin-glycosyltransferase (CGT). The activity of CGT can be confirmed by the hydrolysis of starch on agar plates. In strains like RZ21, RS631 (Lunn Charles et al. 1989) UV radiation is used for mutagenesis instead of chemical reagents, and selection is done by means of T7 bacteriophage. Another way to recognise strains that leak RNase I is to plate them in

Table 1 The bacterial strains developed for allowing efficient release of periplasmic proteins

Bacterial strains	Mutation	Function	References
DS410	minA and/or minB mutant	Enhanced protein secretion	(Böck August et al. 1989)
RZ21, RS631, RL7	Recombinant Vector Transformation using pRGT857-11	Specialized – resistance to infection from T7 phages; enhanced periplasmic secretion protein.	(Lunn Charles et al. 1989)
DS410, BW7261, WCM 100, WCM88	Recombinant Vector Transformation – lambda phage or plasmid vectors	Hirudin derivative for anti-coagulation therapy	(Schmid Gerhard and Habermann Paul 1991; Dassler Dr and Wich Dr 2006)
Other Strains	Using Bacteriocin Release Proteins – also called lysis proteins.	Might enhance secretion but induces leaks of cellular components.	(Pugsley 1983; Lührink et al. 1989; van der Wal et al. 1995)
L-Form Cells	Extreme Alteration/ Complete Absence of peptidoglycan cell wall	Used for secreting various proteins. Cells, though invulnerable to phages, are not structurally sound and exhibit an altered cellular shape.	(Gumpert et al. 1996; Gumpert and Hoischen 1998; Hoischen et al. 2001)

agar containing RNA from yeast. These can be recognised visually by colonies surrounded by halos. Repeating mutagenesis can lead to improving strain development. Now a days, genomic engineering has become the go-to tool for the development of mutant strains instead of the overt reliance on chemical or physical methods. Usage of gene knockouts (Sharma et al. 2020), phage transduction (Eugen et al. 2022), recombinase-based systems (Egger et al. 2020), and also, recently CRISPR-Cas9 mediated mutations (Tong et al. 2021; Shukal et al. 2022). Usage of phage transduction can allow the mutation transfer from one *E. coli* cell with the desired mutation to another cell/strain (Thomason et al. 2007; Tiruvadi Krishnan et al. 2015). Another way to generate *E. coli* mutants is to delete two genes namely *d*, *d*-carboxypeptidase A and B (*dacA* and *dacB*) leading to the disturbance of the cell wall peptidoglycan network and its weakening. This mutation increases the accumulation of soluble intracellular peptidoglycan and changes the cell shape and appearance. The mutation of *dacA* and *dacB* affects the rigid nature of the cell wall. Expression and secretion of extracellular proteins like collagen E4 and fibroblast growth factor receptor 2 (FGFR2) can be done by this method (Yang et al. 2018). The usage of CRISPR-Cas9 is recent and the success of this approach remains challenged by the time. However, the following strains mentioned below, and their corresponding mutations have been proved excellent choices for secretion of desired proteins. Though these strains proved effective, their efficiency differ from one another in terms of production and secretion. The list of successful strains is mentioned here: WCM88 (Schmid Gerhard and Habermann Paul 1991), WCM100 (Dassler Dr and Wich Dr 2006), *lpp* mutants (Yem and Wu 1978), *lpp mrcA* double mutants (Chen et al. 2014), mutants of *pal* (Lazzaroni et al. 1989), *mrcA* (Chen et al. 2014), and *mrcB* (Natarajan et al. 2017). CRISPR-Cas9 technology has revolutionized bacterial genome engineering, offering versatile applications beyond enhancing membrane permeability for secretion. In the context of recombinant protein production, CRISPR-Cas9 enables precise gene knockouts

or insertions to regulate the expression of secretion-related genes, such as those encoding chaperones, signal peptides, or outer membrane transporters (Jiang et al. 2013). Additionally, it facilitates strain optimization by reducing protease activity, minimizing protein degradation, and improving cellular stress responses, thereby increasing overall secretion efficiency (Ping et al. 2023; Agrawal et al. 2024). The ability to fine-tune metabolic pathways and control regulatory elements through CRISPR-based modulation further enhances its potential for optimizing bacterial secretion systems, making it a powerful tool in synthetic biology and biopharmaceutical applications.

Increasing secretion of target protein of interest by mutagenesis of osmotically inducible protein Y (OsmY)

Usage of mutated signal peptides and secretory carrier proteins is often done to enhance production of recombinant proteins in Gram-negative bacteria. The *E. coli*'s OsmY (Osmotically Inducible Protein Y) (Gonzalez-Perez et al. 2021b) secretes the proteins extracellularly. By applying random and site-specific mutagenesis by recombinase method in the Bacterial Extracellular Protein Secretory System (BENNY), it was identified as the best carrier protein. Combination mutagenesis has identified OsmY (M3) (L6P, V43A, S154R, V191E) as the best carrier protein as this has shown to improve secretory production of Tfu0937 β -glucosidase protein (Gupta et al. 2013). Other proteins that have been produced successfully through this method include β -glucosidase, endoglucanase (Gupta et al. 2013), xylosidase, and xylanase (Le and Wang 2014). The carrier protein OsmY modulation has also exhibited increased activity and proper folding of highly unstable chaperone proteins (Le and Wang 2014; Patwa and Deep 2022).

Inducing permeability for enhancing protein secretion

For the secretion of recombinant proteins into the production medium, membrane permeability can also be induced

during cultivation by specific protein co-expression. The most widely used proteins for this purpose is BRP (Bacteriocin Release Proteins), also called kill or lysis proteins (van der Wal et al. 1995; Sommer et al. 2010). The most recognised BRPs include colicin A (van der Wal et al. 1995), E1 and E2 (Pugsley 1983), as well as cloacin DF13 (Dufour et al. 2017). However, it has been observed that the outcome of using BRP on secretion of target proteins is non-specific and might result in the leakage of cytoplasmic and periplasmic components. Other issues of using this method of BRP-co-expression are quasi-lysis and lethality, resulting in decreased cell numbers, decreased growth, and decreased productivity. Therefore, this strategy requires strict control of the use of BRP co-expression (Beshay et al. 2007).

Chemical methods for enhanced protein secretion

Chemical methods to enhance protein production are critical in optimizing yields for industrial and therapeutic applications. One of the primary approaches is media optimization, which involves tailoring the composition of the growth medium to support maximal protein expression. This includes adjusting carbon and nitrogen sources, buffering agents, salts, vitamins, and trace elements to suit the metabolic needs of the host organism. For instance, the inclusion of glucose or glycerol as carbon sources can provide an energy boost, while the addition of amino acids or peptones reduces the metabolic burden of synthesizing these precursors de novo, allowing the host to channel resources toward protein production (Zhang et al. 2022). Media pH and osmolarity also play significant roles as deviation from these factors can stress the host cell and reduce protein yield. Chemical additives like inducers (e.g., IPTG or arabinose) are widely used in inducible expression systems to trigger protein synthesis, while certain surfactants, such as Tween-20, can reduce aggregation of hydrophobic proteins (Singh et al. 2017). Another strategy is the inclusion of chemical chaperones like DMSO, glycerol, or betaine, which stabilize folding intermediates and reduce misfolding or aggregation (Patwa and Deep 2022). Similarly, redox modulators like cysteine, glutathione, or dithiothreitol can enhance disulfide bond formation, particularly in periplasmic expression system (Knoke et al. 2023).

Moreover, metal ion supplementation (e.g., Mg^{2+} , Ca^{2+} , Zn^{2+}) can enhance protein stability or function, especially for enzymes and metalloproteins, while chelators like EDTA are occasionally added to minimize the impact of harmful metal ions (Rosano and Ceccarelli 2014; Shalev 2022). The addition of antioxidants like ascorbic acid or thiourea can reduce oxidative stress in host cells, improving cell viability and protein yield. Furthermore, auto-induction media which use lactose, or its derivatives can enable a gradual and

sustained induction phase, reducing the metabolic shock often caused by abrupt inducer addition (Faust et al. 2015). Beyond the medium itself, feeding strategies such as fed-batch fermentation allow the gradual addition of nutrients, minimizing substrate inhibition and toxic byproduct accumulation (Bolmanis et al. 2023). Additionally, supplementing the media with growth factors, nucleotides, or cofactors like NAD^+ can support cellular growth and recombinant protein synthesis. Chemical inhibitors of proteases, such as PMSF or leupeptin, are frequently used to prevent degradation of the target protein during or after expression (Ryan and Henahan 2017).

An extreme example of cell wall alteration: the L-form cell

In other type of cells called L-form cells, the peptidoglycan cell wall is extremely altered or completely absent, thereby completely changing cellular morphology of the bacteria. Though the lack of cell walls renders the bacterial cells invulnerable to bacteriophages or antibiotics (Mercier et al. 2014; Aleksandra et al. 2022). There are several drawbacks of using L-form cells for protein production. Due to their altered cellular structure, the bacteria grow slower and lack cellular robustness, making them physically weak and unfit for industrial use as they are vulnerable to environmental changes and other chemicals like TritonX-100 (Gumpert and Hoischen 1998). Though some proteins like penicillin G-acylase have used the L-form cell for their production, but yield was seen to be comparatively low (Gumpert et al. 1996). Therefore, researchers have started to move away from the use of L-form cells for their low-yields and difficult handling, favouring other methods for production of proteins of interest instead.

ESETEC®

Developed by the Wacker Company, ESETEC® is an *E. coli*-based protein secretion system (Fig. 3a). This commercial technology platform is based on the *E. coli* K12 strain (Gupta and Shukla 2017) and is genetically deletion of genes *minA* and *minB* in outer membrane space to possess a secretion of recombinant proteins after the expression (Böck August et al. 1989). Along with the above modifications, the strain also possesses alterations in the outer membrane (OM) and signal peptides and therefore is capable of secreting proteins at a larger scale in a fermenter (Fig. 3a). This large-scale production capability allows this production system to be very versatile i.e. it can produce non-glycosylated proteins of various origins – prokaryotic, eukaryotic and mammalian; with a wide range of sizes (between 5kD and 150kD). ESETEC® uses plasmids with a variety of origin of

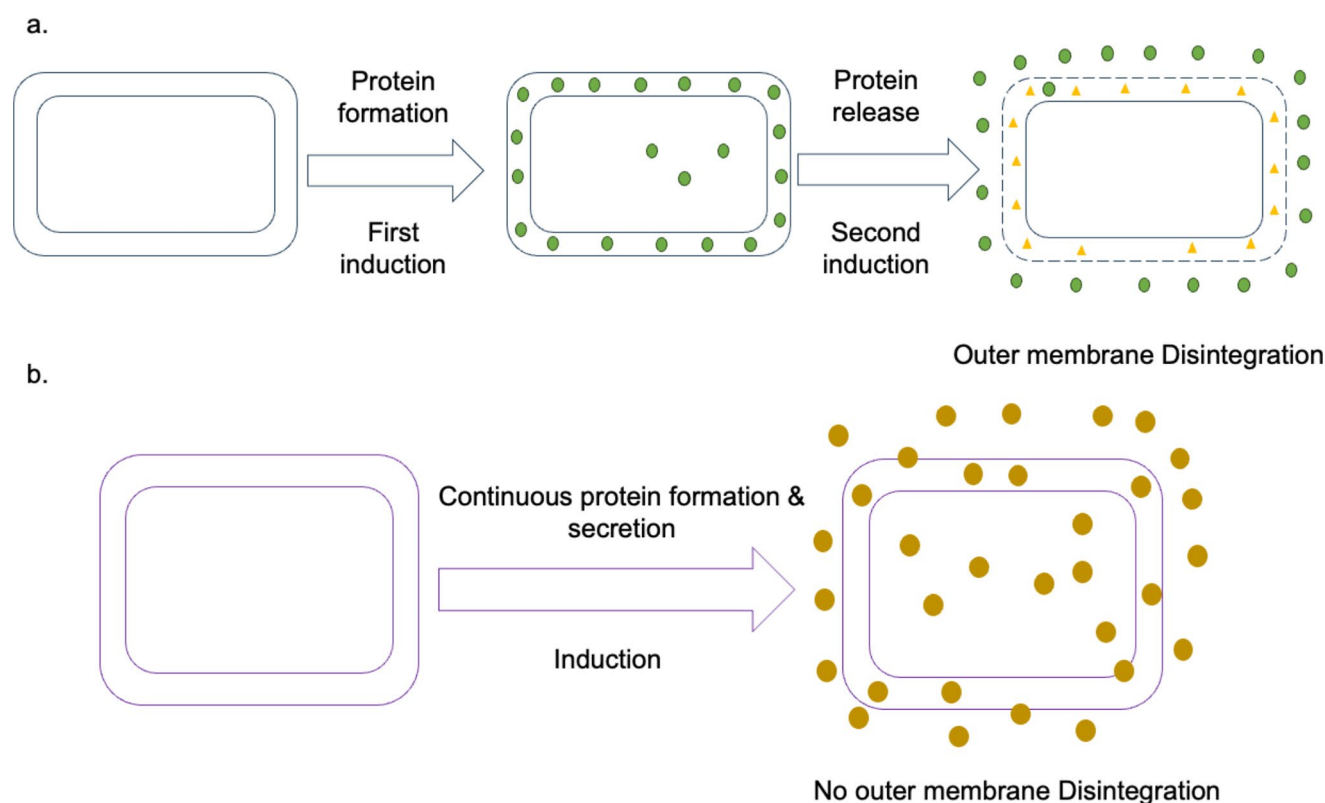


Fig. 3 (a) ESETEC® by Wacker. The first induction of the culture allows enhancement of the production of the protein (in green) and its release into the periplasm while the addition of the second induction using a release factor (in yellow) allows the extracellular release of

the accumulated protein. (b) BacSec® by Oncosimis Biotech. A relatively recent proprietary technology, BacSec® allows for the continuous secretion of the desired product as it is produced, allowing for the reduction of manufacturing costs

replication (*ori*'s) for controlled expression, *tac* promoter, and the *lacIq* repressor for induced expression. Due to the success of this method, Wacker upgraded to ESETEC® 2.0. The upgrade involves more novel plasmids with unique signal peptides and also tools for co-expression of chaperones, di-sulphide bridges forming co-factors, and the Sec-pathway apparatus to increase yield of protein of interest. This has improved the yield of proteins by about four times as described in their website.

BacSec® (bacterial secretory)

Developed by Oncosimis Biotech, BacSec® is a relatively recent proprietary technology (Fig. 3b). It was developed to address the typical problems faced during mass production of proteins using bioreactor-based methods without disintegration of outer membrane. This technology relay on *E. coli* strains (mutated transporter system), media composition, expression vector and signal peptides. The BacSec® technology makes it easy by integrating both the upstream and downstream processes in a continuous bioprocessing mode (Fig. 3b) (Lokireddy 2020a; Lokireddy and Kona 2020). Utilization of the perfusion-capable bioreactor in

conjunction with BacSec® technology addresses the limitations such as deteriorating environment by addition of toxic proteins, protein aggregation as result of protein crowd, large footprint by reactors, limited dilution rates, termination of cycle time, removal by-products, and limited productivity per day. Apart from that, decrease in inhibitory levels of product concentration to *E. coli* posed during fed-batch or batch methodologies are also addressed (Lokireddy 2020a; Lokireddy and Kona 2020). Therefore, applying BacSec® and perfusion technologies for continuous production of biologics results in 5 to 10-fold higher yield compared to conventional methods with significant reduction in manufacturing cost. Hence, it is necessary to establish perfusion-capable bioreactor in conjunction with BacSec® technology to increase yield of biologic production and decrease the costs of manufacturing simultaneously. The use of BacSec® enables secretion of bacterially expressed recombinant protein (non-glycosylated) through the fermentation process, as well. It enables purification of secreted protein through simple downstream processing with negligible contamination from bacterial endotoxins, reducing further costs for purification. Preliminary data published in patents shows that several biologics are effective as counter parts available

in the market in a laboratory setting and that it has a high potential for a high-scale production within the biopharmaceutical industry (Lokireddy 2020a, b, c; Lokireddy and Kona 2020).

ESETEC[®] and BacSec[®] are both advanced secretion systems designed to enhance recombinant protein production on a commercial scale, but they differ in their underlying approaches and applications. ESETEC[®], developed by Wacker, is based on engineered *Escherichia coli* strains where minA and minB genes deleted that enable direct extracellular secretion of recombinant proteins via optimized Sec pathway components, reducing periplasmic accumulation and thus simplifying purification. In contrast, BacSec[®], a technology from Oncosimis Biotech, integrates continuous bioprocessing with engineered secretion mechanisms to enhance yield and streamline production while minimizing host cell protein contamination without disintegrating the outer membrane. While ESETEC[®] focuses on efficient secretion through two step induction by disintegration of outer membrane, BacSec[®] is designed for scalable, cost-effective biomanufacturing with a focus on continuous production through perfusion method without disintegration of outer membrane. Together, these technologies represent complementary approaches to improving secretion and process economics in industrial biotechnology.

Future prospects and concluding remarks

The production of recombinant proteins has come a long way since their advent in the 1970s by the use of Restriction Endonucleases. This innovation led to the efforts by Eli Lilly and Co. that brought recombinant human insulin to the doorsteps of millions of diabetic patients who did not have to suffer with anaphylaxis triggered by their pig or bovine counterparts. The ability to genetically manipulate has gifted an industrial boom of diversified products that have been used in the medical, pharmaceutical, and dairy industries to name a few. Many other methods for the manipulation of DNA have been devised – the use of vectors, phage recombination systems. Though it has become more sophisticated with time, many hurdles have consequently appeared due to the systemic issues that prokaryotes present due to the limited cellular capabilities, despite their several advantages. With the advances in genetic engineering, it has become easier to manipulate the organism *E. coli*, which, by all accounts, is a poor secretor, to become a moderately good secretor by making the outer membrane more permeable through induced co-expression of various secretory systems and the modifications of the fermentation media to improve yields of proteins of interest.

The future of recombinant protein production is shifting from full monoclonal antibodies to antibody fragments

and peptides, which offer higher specificity and simpler manufacturing processes. Secretory systems must address challenges such as aggregation, folding efficiency, and scalability to realize their full potential in industrial applications. Future innovations aim to address these issues through a combination of genetic engineering, synthetic biology, and process optimization. One promising approach is the development of engineered chaperone systems that assist in proper protein folding, minimizing misfolding and aggregation. Additionally, advancements in secretion signal engineering, including the rational design of hybrid signal peptides, could enhance translocation efficiency and secretion yield. Another critical area of research is the integration of CRISPR-Cas9 for precise strain modifications, allowing for targeted gene regulation to optimize secretion pathways while minimizing proteolysis and cellular stress. To further improve scalability, continuous bioprocessing methods, such as perfusion-based fermentation, are gaining industrial attention. These systems enable steady-state production, reducing batch-to-batch variability and increasing overall yield. Moreover, computational modeling and artificial intelligence-driven strain optimization are emerging as valuable tools to predict and enhance bacterial secretion capabilities. The combination of these strategies will enable more efficient, cost-effective, and scalable recombinant protein production. Due to the importance of this technology, improvements will continue to happen in the future because the use of *Escherichia coli* is ubiquitous in industries and laboratories all around the world due to its ease of use and safety in application. The interest in *E. coli* biology is not just limited to fundamental research, but also the field of translational research as many of its facets are yet unknown to us. The discovery of a new biological pathway, for example, may lead to the development of a new secretory system. Therefore, the future for this organism still looks bright and promising.

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